The role of remote Photoplethysmography in Musical Biofeedback Systems for Stress Reduction and Consciousness Regulation

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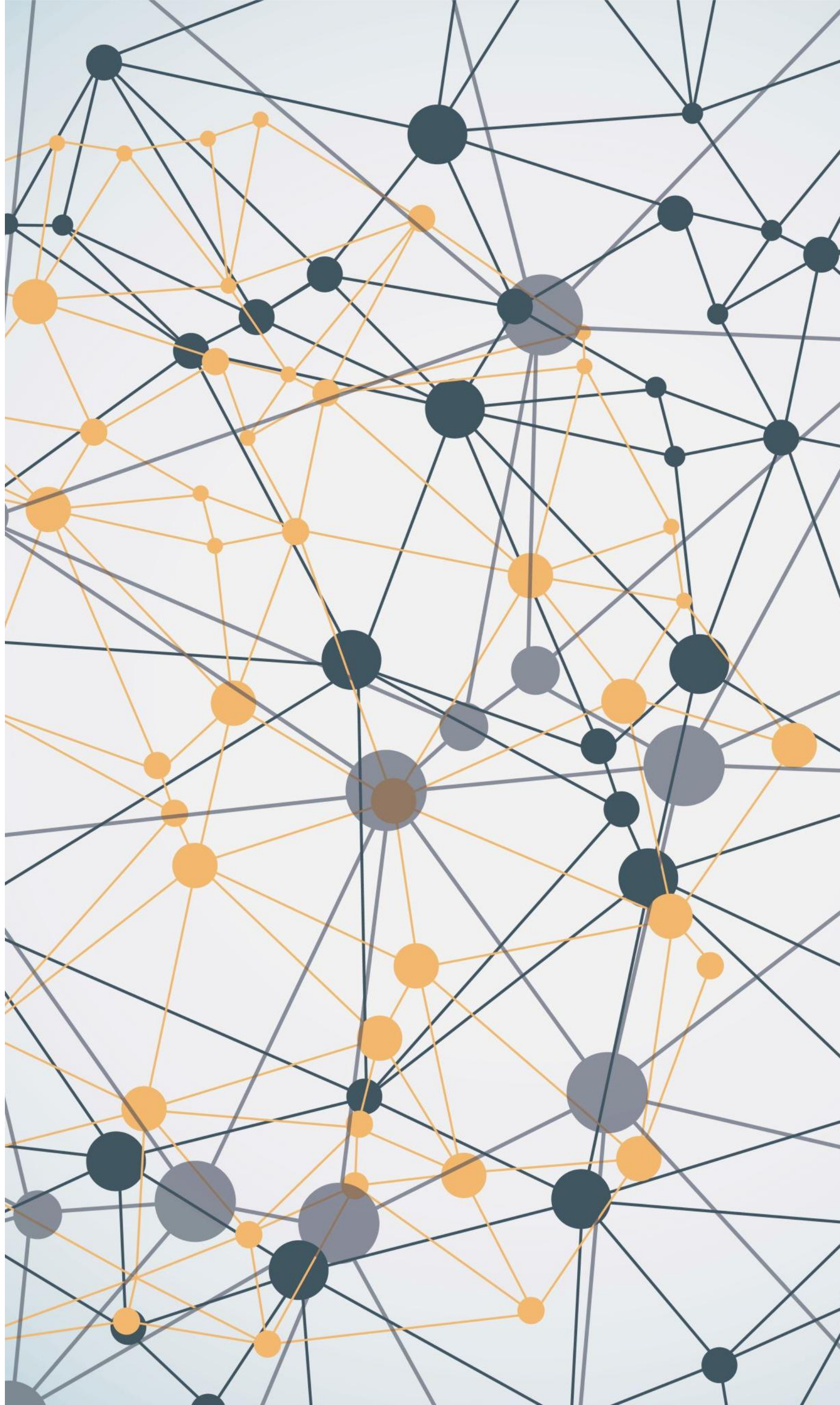

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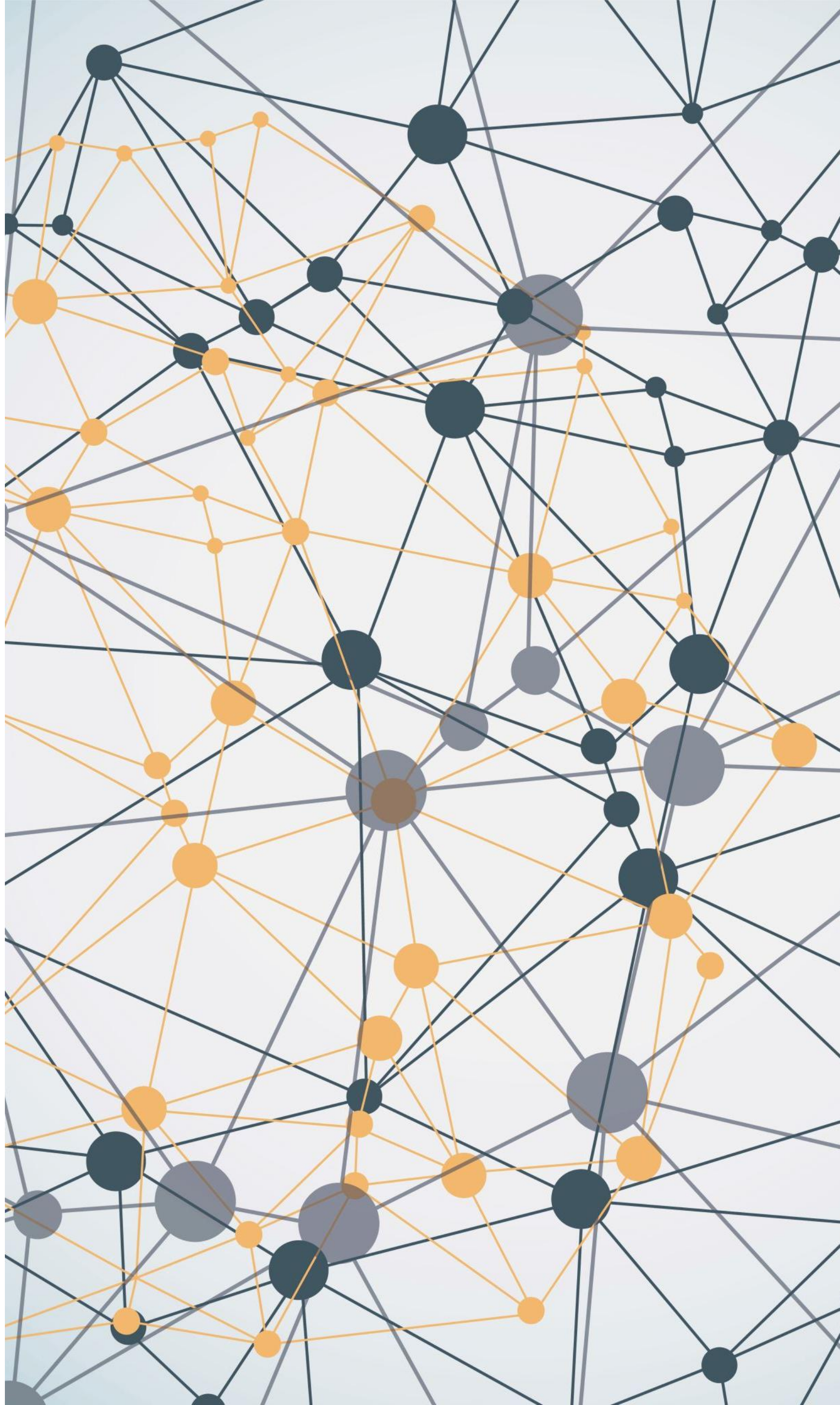


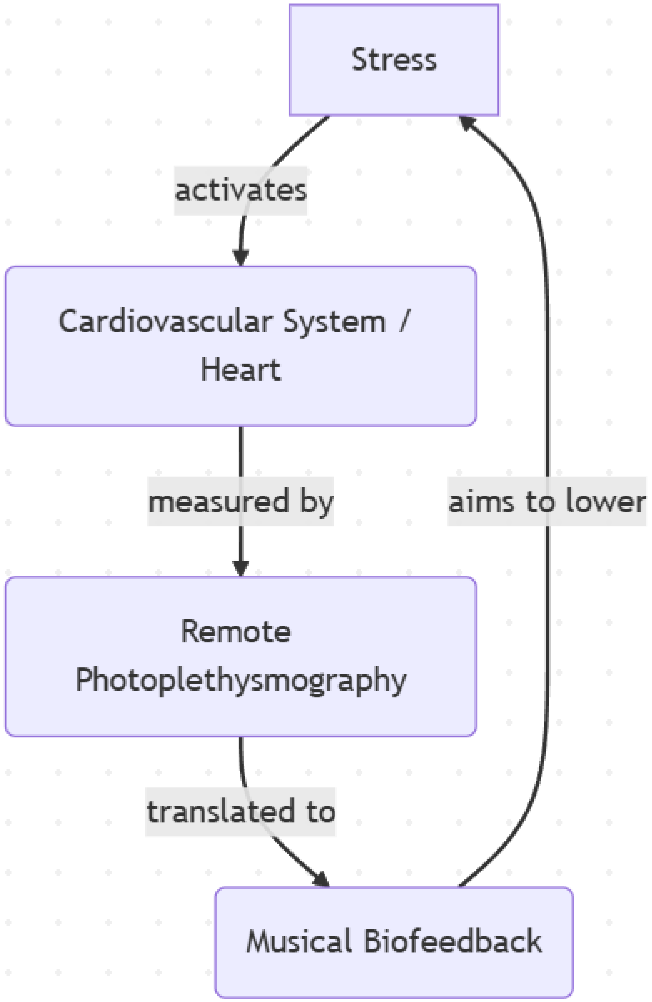
Introduction

- Stress changes the heart rate[1]

- Plethysmography: Measurement of pulse waves[2]
- Photoplethysmography (PPG): … using light reflectance[2]
- remote PPG (rPPG): … via digital camera (contactless)[2]

- Biofeedback: Real-time loop returning physiological data[3]
- Musical Biofeedback: Auditory feedback loop using music[3]

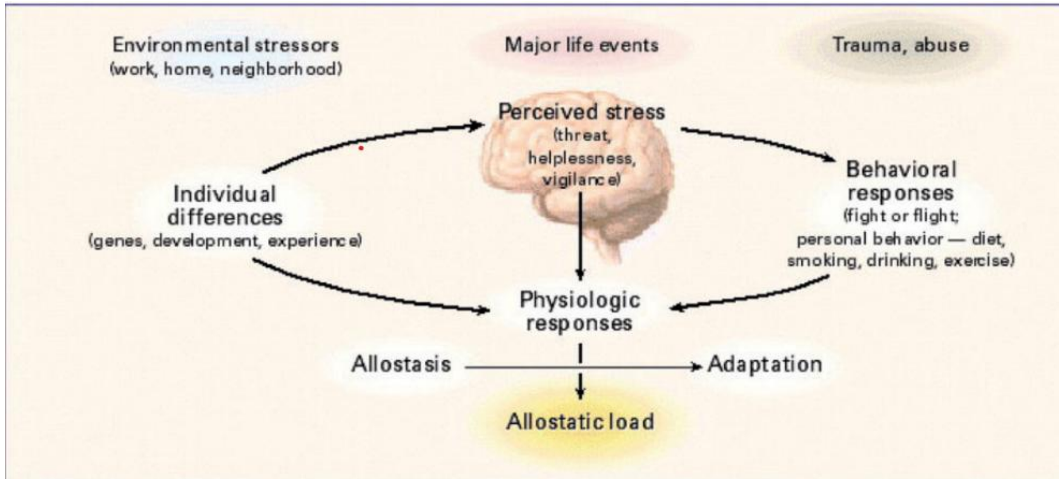
[1] cf. Mcewen2017neurobiology, [2] cf. verkruysse2008remote, [3] cf. giggins2013biofeedback



Musical Biofeedback Loop: Real-time translation of physiological rPPG data into intuitive musical biofeedback
Source: Own illustration

Stress

- Protective Response: Short-term mobilization is a vital survival mechanism[1]
- Maldapative Coping: Harmless situations trigger stress, leading to harmful habits (smoking, drinking)[1]
- Heart Rate (HR): Beats per minute – represents the quantity of heart activity[2]
- Heart Rate Variability (HRV): Time variation between consecutive beats – represents the quality of heart activity[2]



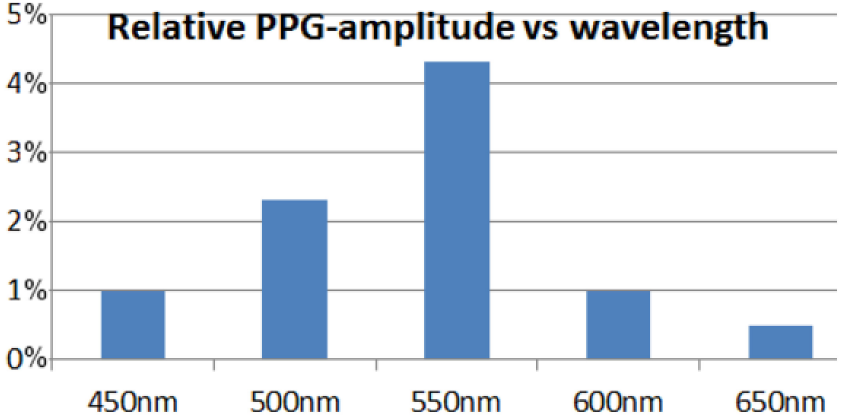
The figure illustrates the separate components that the brain takes into account when releasing stress[1]

[1] cf. Mcewen2017neurobiology [2] cf. shaffer2017overview

Remote Photoplethysmography (foundation)

- Absorption: Blood volume changes modulate light absorption with each beat[2]
- Green Channel: Provides the strongest signal[2]

- Target Areas: Cheeks offer the highest signal-to-noise ratio: easily accessible, exposed[3]



Spectral sensitivity of the PPG signal. Green light exhibits the highest amplitude, making it the most reliable channel for non-contact heart rate monitoring.[1]

The quality of the rPPG signal in the face. Blue shows low activity and red high activity[3]

[1] cf. dehaan2013chrom, [2] verkruysse2008remote,
[3] cf. lempe2013roi

Remote Photoplethysmography (Signal extraction overview)

- Blind Source Separation
  - Separation of pulse and noise based on statistical independence[1]

- Physics based Models (CHROM)
  - Principle: uses skin-light-reflection properties for robust real-time tracking[2]

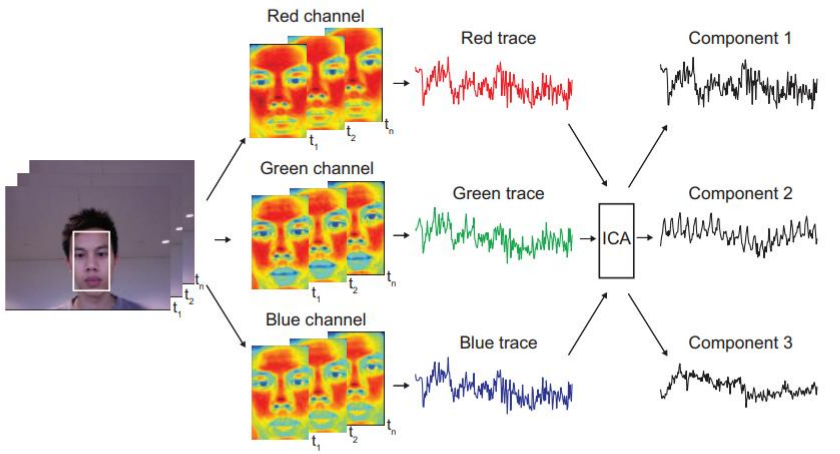
- Deep Learning Models
  - Principle: Uses neural networks high accuracy[3]

[1] cf. poh2010non, [2] cf. dehaan2013chrom [3] cf. Debnath25

Remote Photoplethysmography (Blind Source Separation)

- Signal Mixture: the camera image as a mixture of independent sources (Pulse, Light, Motion)[1]
- Mathematical Tool: Uses algorithms like JADE so separate the signals[1]

- Signal Refinement (Filtering)[1]
  - Bandpass Limit: Restrict data to 45-240BPM
  - Fast Four Transform Analysis: find the strongest periodic frequency (the Heart rate)
  - Outlier Rejection: Prevents „unrealistic jumps“
- Limitation[1]:
  - Component Misidentification: In extreme light or motion, the wrong signal (light instead of pulse) is picked
  - Requires a long data window (30 seconds)



Separating independent sources from raw skin color channels (RGB) using the JADE algorithm for rPPG[1]

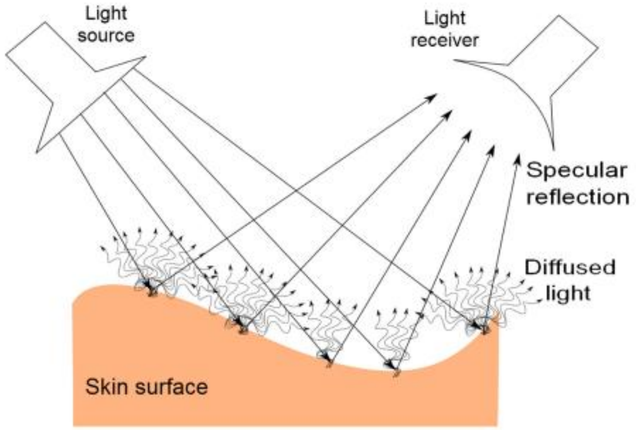
[1] cf. poh2010non

Remote Photoplethysmography (CHROM)

- Physicsbased alternative: Chrominance-based method[1]

- Camera sees:[1]
  - Intensity: Brightness of light
  - Specular Reflection: Mirror like glare on the surface
  - Static Skin Color: The constant tissue color
- 2D-Projection RGB signals are projects onto a plane perpendicular to the standard skin tone [1]
- Noise cancellation: A clean, movement-robust pulse signal[1]

- Bandpass Filtering (0.7 – 4.0Hz) -> matching the human heart rate (42-240BPM)[1]
- Fast Fourier Transform: identifies the strongest periodic signal to determine the heart rate[1]



The illustration depicts light reflection on the skin [1]

[1] cf. dehaan2013chrom

Remote Photoplethysmography (CHROM)

- Accuracy 98% in stable conditions, 92% correleation with medical sensors (ground truth)[1]
- The "Motion" Advantage: Maintains a 48% success rate under heavy movement - where BSS models drop to a failing 4 – 11%[1]
- Low latency: Only 1.6s (vs 30s with BSS)[1]
- Diversity: Robust performance across all skin tones[1]

[1] cf. dehaan2013chrom

Remote Photoplethysmography (Deep Learning Models)

- Transition to End-to-End Deep Learning Models[1]
- Fully automated: No manual Face/ROI selection required[1]

- Data challenges:[1]
  - Significant lack of large-scale, high-quality datasets
  - Underrepresentation of diverse age groups and skin pigmentations
- Mobile Optimization: Ongoing research into lightweight models for real-time deployment on smartphone hardware[1]

[1] cf. Debnath25

Musical Biofeedback

- Ecological Perception:[1]
  - Uses natural associations (e. g. distance) to represent intensity
  - Example: Decreasing volume mimics a sound source moving away, making data changes intuitive
- Data-to-Music Transformation[1]
  - direct transformation of heartbeat data into an auditory stream
  - Continuous parameter mapping: ensures immediate, fluid feedback of physiological changes

[1] cf. dubus2013systematic

Regeneration and Consciousness Regulation

- Cardiovascular Synchronization[1]
  - Resonance Frequency: Achieved at ~6 breaths per minute
  - Harmonic State: Heart rate and blood pressure synchronize for maximum efficiency
  - Optimized gas exchange in the lungs
  - Less strain on the cardiovascular system

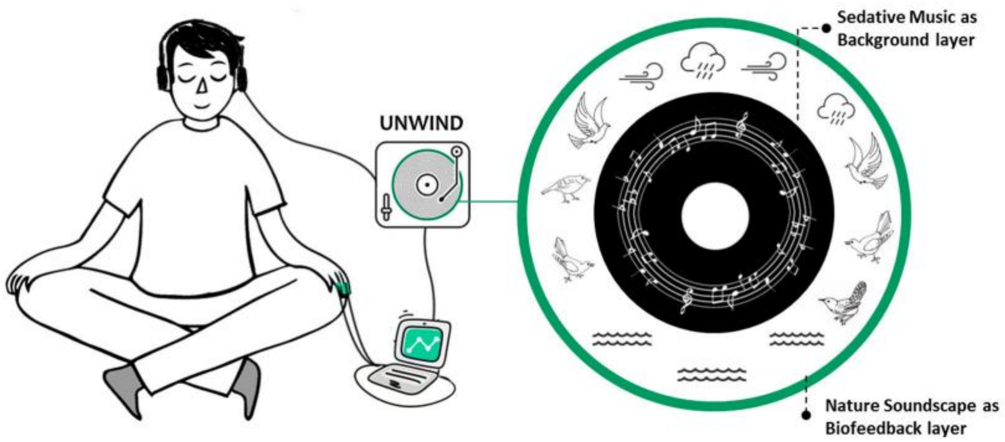
- Physiological Impact[1]
  - Deep Relaxation Response in the body
  - The auditory signal acts as a meditative anchor

- Mindfulness[2]
  - Enhances sensitivity to thoughts and emotional states
  - Promotes an attitude of curiosity, openness and acceptance
  - Helps to perceive the internal state without judgement

[1] cf. lehrer2014heart, [2] cf. bishop2004mindfulness

Case Studies (Unwind System)

- Concept: Nature soundscapes with sedative music to guide relaxation[1]
- HRV Sonification[1]:
    - Wind Mapping: Heart Rate Variability (HRV) is mapped to wind sounds
    - Users adapt their breathing to transform the wind into a smooth, consistent sine wave
- Adaptive Soundscape[1]:
    - LOW HRV (Stress): Environment becomes "busy" and complex
    - High HRV (Relaxed): Soundscape becomes simple, transparent and calm

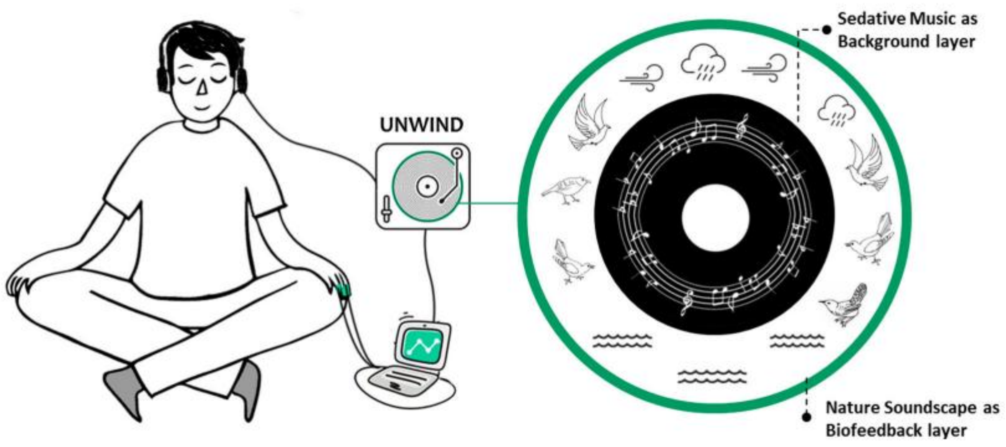


The illustration depicts the feedback loop of the Unwind System[1]

[1] cf. yu2018unwind

Case Studies (Unwind System)

- Key Results:[1]
    - Significant reduction in both heart rate and respiration rate in comparison to the control group
    - 70% of participants used the nature sounds to visualize scenes, significantly improving focus and immersion.



The illustration depicts the feedback loop of the Unwind System[1]

[1] cf. yu2018unwind

Case Studies (Muse™ Headband)

- The concept: Lowering the entry barrier for meditation through real-time brainwave measurement[1]
- Daily attention training with feedback through smartphone app[1]

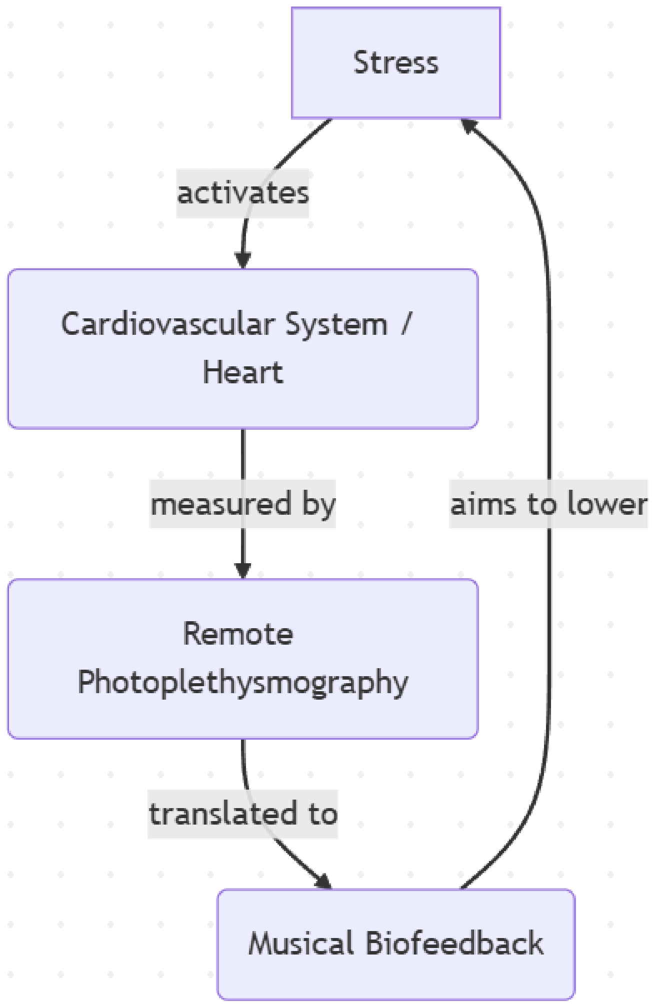
- Key Findings:[1]
    - Stress Reduction: The experimental group recorded a 7% decrease after 2 weeks 14% after 4 weeks
    - Mental Performance: Improved concentration levels and superior impulse control during complex tasks
    - Control Group: Showed no significant changes in stress or performance levels
- Real-time feedback is the decisive factor[1]

Subject wearing the Muse™ EEG headband during a neurofeedback session.[1]

[1] cf. app7121280

Conclusion

- The Core Metric: Heart Rate Variability (HRV)[1]
  - HRV provides significantly deeper physiological insights than simple Heart Rate (BPM)
- Technical Strategy: [2]
  - Algorithm: Choosing CHROM for its computational efficiency and real-time capabilities
- The Competitive Edge:
  - Unlike Blind Source Separation[3], CHROM requires shorter observation windows and effectively identifies the correct color channels [2]
  - Superior to Deep Learning: CHROM is robust against noise without the massive resource intensity or dataset bias of AI models[4]



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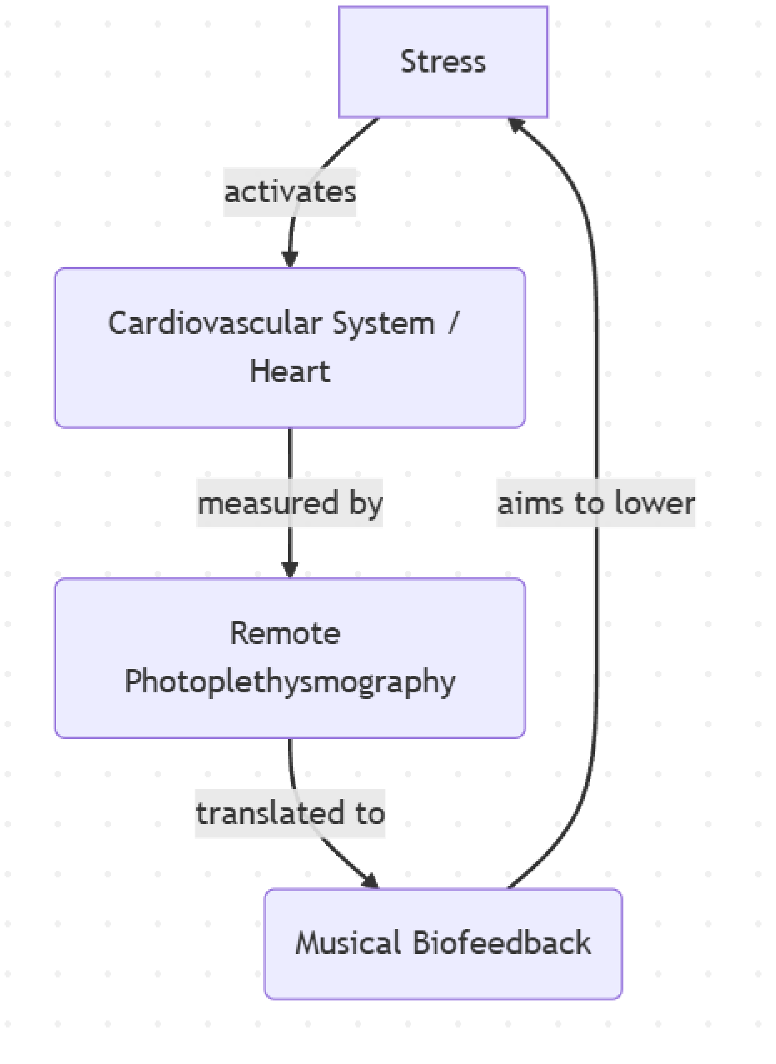
[1] cf. shaffer2017overview, [2] dehaan2013chrom, [3] cf. poh2010non, [4] cf. Debnath25

Conclusion

- The mapping Process[1]:
  - Continuous Mapping: instant, fluid audio stream
  - Sound Design: “Calm State = Calm Sound” – non intrusive soundscapes to reinforce relaxation
- Evidence-Based Systems:
  - Unwind System**:** uses nature sounds for feedback[2]
  - Muse™**:** uses real-time loops to increase stress awareness and control[3]



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[1] cf. dubus2013systematic [2] cf. yu2018unwind [3] cf. app7121280

Future Outlook

- Potential: Monitor stress at work (office)
- Trend: Small AI models for laptops or mobiles
- Efficiency: Cost –effective (simple camera)
- Privacy: High risk! Local processing only (No Cloud)

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